§12.5 Directional derivatives & gradients.

Recall $\quad f_x(\bar{P}) = \lim\limits_{h\to 0} \dfrac{f(\bar{P}+h\,i)-f(\bar{P})}{h}$ $\qquad \bar{P}=(x,y)$

$i=(1,0)$

$j=(0,1)$

$f_y(\bar{P}) = \lim\limits_{h\to 0} \dfrac{f(\bar{P}+h\,j)-f(\bar{P})}{h}$

If $\bar{u}$ is a unit vector, then we define the directional derivative of $f(x,y)$ at $\bar{P}$ in the direction $\bar{u}$ by

$$D_{\bar{u}} f(\bar{P}) = \lim_{h\to 0} \frac{f(\bar{P}+h\bar{u})-f(\bar{P})}{h} \qquad \bar{u}=(u_1,u_2)$$

This is the derivative of $g(h) = f(\bar{P}+h\bar{u})$ at $h=0$;

Theorem If $f(x,y)$ is differentiable at $\bar{P}$, then

$$D_{\bar{u}} f(\bar{P}) = \bar{u}\cdot\nabla f(P) = u_1 f_x(x,y) + u_2 f_y(x,y)$$

Proof $\quad \dfrac{f(\bar{P}+h\bar{u})-f(\bar{P})}{h} = \dfrac{\nabla f(\bar{P})\cdot(\cancel{h}\bar{u}) + \epsilon(h\bar{u})\cdot\cancel{h}\bar{u}}{\cancel{h}}$

$$= \nabla f(\bar{P})\cdot\bar{u} + \underbrace{\epsilon(h\bar{u})\cdot\bar{u}}_{\downarrow \text{ as } h\to 0 \atop 0} \qquad \square$$

Ex $\quad f(x,y) = xy \quad$ Compute its derivative at $p=(2,3)$ in direction of $(4,3)$.

$\nabla f = (y,x) \quad \nabla f(2,3) = (3,2) \quad u = \dfrac{1}{\sqrt{16+9}}(4,3) = (4/5, 3/5)$

$D_u f(2,3) = (4/5, 3/5)\cdot(3,2) = 18/5$

Ex $f(x,y,z) = \frac{xy}{z}$ at $\bar{p} = (2,-1,1)$ in direction $(1,1,1)$

$$\nabla f = \left(\frac{y}{z}, \frac{x}{z}, \frac{-xy}{z^2}\right) \qquad \nabla f(2,-1,1) = (-1, 2, 2) \qquad \bar{u} = \frac{1}{\sqrt{3}}(1,1,1)$$

$$D_{\bar{u}} f(2,-1,1) = \frac{1}{\sqrt{3}}(1,1,1)\cdot(-1,2,2) = \frac{3}{\sqrt{3}} = \sqrt{3}$$

<u>Remark</u> $D_{\bar{u}} f(\bar{p}) = \bar{u}\cdot\nabla f(\bar{p}) = \|\bar{u}\|\,\|\nabla f(\bar{p})\|\cos\theta$

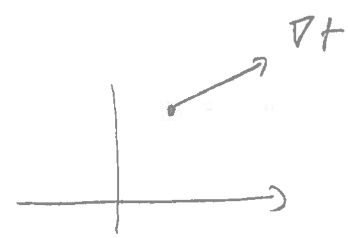
$$= \|\nabla f(\bar{p})\|\cos\theta$$

So $D_{\bar{u}} f(\bar{p}) \leq \|\nabla f(\bar{p})\|$ ← max when $\theta = 0$ ie $\bar{u} = \frac{\nabla f(\bar{p})}{\|\nabla f(\bar{p})\|}$

$\geq -\|\nabla f(\bar{p})\|$ ← min when $\theta = \pi$ ie $\bar{u} = \frac{-\nabla f(\bar{p})}{\|\nabla f(\bar{p})\|}$

$= 0$ if $\theta = \frac{\pi}{2}$ ie $\bar{u}\cdot\nabla f(\bar{p}) = 0$

<u>Eg</u> $f(x,y) = xy \qquad p = (1,2)$



$\nabla f = (y, x) \qquad \nabla f(1,2) = (2,1)$

Max for $D_{\bar{u}} f(1,2)$ is $\sqrt{5}$ in direction $\bar{u} = \frac{1}{\sqrt{5}}(2,1)$

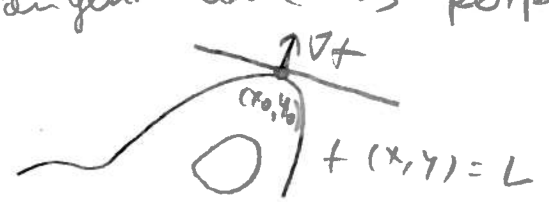
Min " " " $-\sqrt{5}$ " " $\bar{u} = \frac{-1}{\sqrt{5}}(2,1)$

$D_{\bar{u}} f(1,2) = 0$ in direction $\pm\frac{1}{\sqrt{5}}(-2,1)$

Remark $f(x,y) = L$ is a level curve.

The tangent line at (x_0,y_0) is $0 = \nabla f(x_0,y_0)\cdot(x-x_0, y-y_0)$

So the tangent line is perpendicular to ∇f



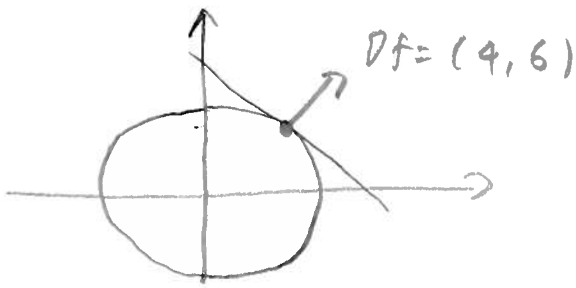
Ex $\quad z = 2x^2+3y^2 \qquad c = 5$

Level curve is $\{5 = 2x^2+3y^2\} = C$

$(1,1) \in C$

$\nabla f(1,1) = (4x, 6y)\big|_{(1,1)} = (4,6)$

Tangent line $\quad 0 = (4,6)\cdot(x-1, y-1) = 4x+6y-10$



Ex The temperature in the region is $T = (50-z)(1-x^2-y^2)$. In what direction does the temperature decrease fastest at the point $(1,1,0)$?

$\nabla T = \big((50-z)(-2x),\ (50-z)(-2y),\ x^2+y^2-1\big)$

$\nabla T(1,1,0) = (-100, -100, 1)$

$u = \dfrac{1}{\sqrt{20{,}001}}(-100, -100, 1)$

Exercises

1) Compute $D_{\bar{u}} f(\bar{p})$ for ($\bar{u}$ is in the direction of $\bar{u}$)

a) $f(x,y) = x^2 y$ $\bar{p} = (1,2)$ $\bar{a} = (3,-4)$

b) $f(x,y) = y^2 \ln x$ $p = (1,4)$ $\bar{a} = i - j$

c) $f(x,y) = e^x \sin y$ $p = (0, \pi/4)$ $\bar{a} = i + \sqrt{3}\, j$

2) Find the direction (unit vector) in which f increases the fastest at $\bar{p}$. What is the rate of change in that direction?

a) $f(x,y) = x^3 - y^5$ $p = (2,1)$

b) $f(x,y,z) = x e^{yz}$ $p = (2,0,-4)$

c) $\sin(3x - y)$ at $p = (\pi/6, \pi/4)$

3) Repeat 2) looking for the direction in which f decreases the fastest.

4) The temperature is given by $T = \dfrac{200}{1+x^2+y^2+z^2}$

Where is it hottest?

In what direction is the greatest increase at $(1,-1,1)$?